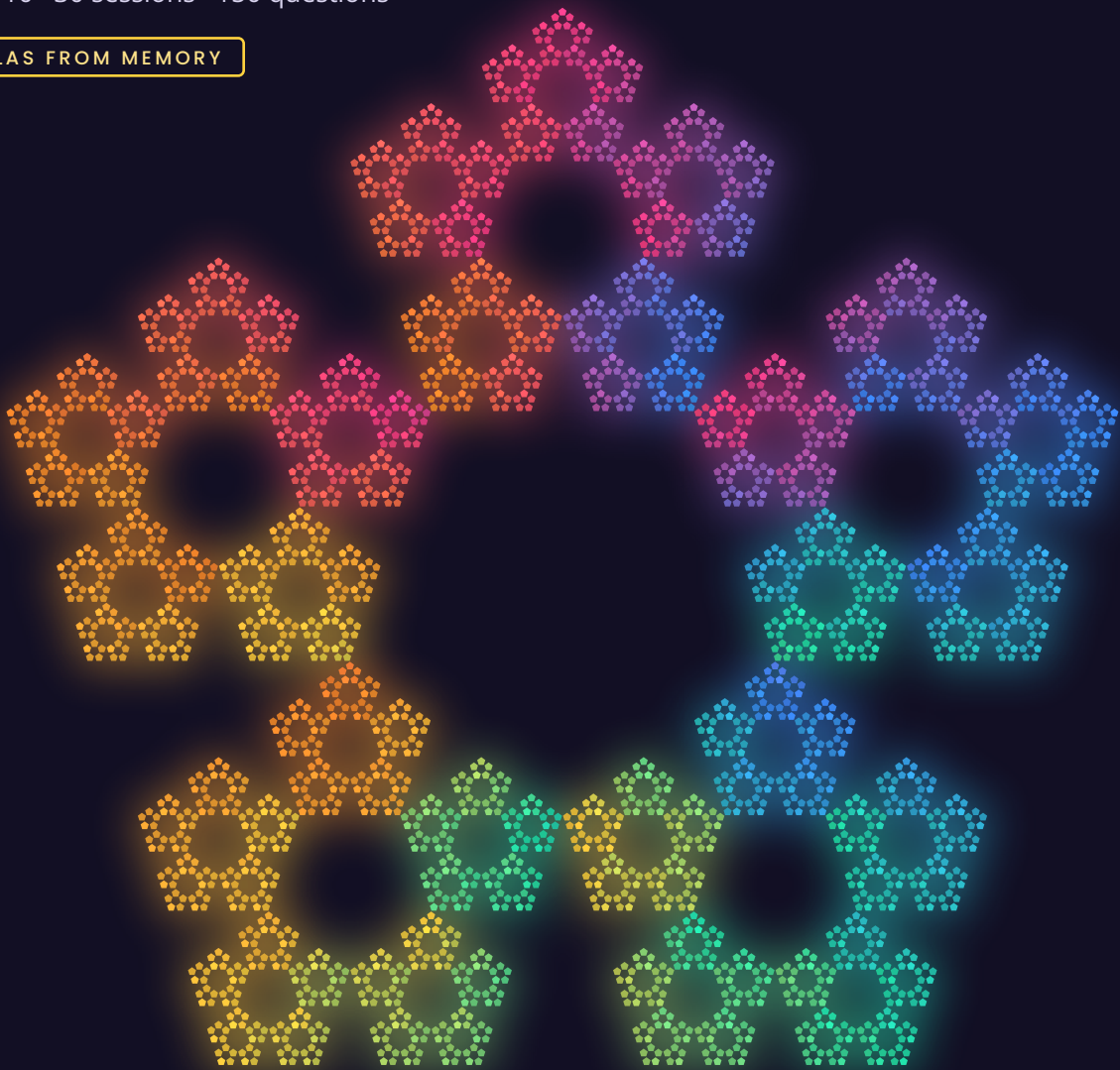


FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS FROM MEMORY



NAME _____

CLASS _____

SIERPIŃSKI PENTAGON

five copies of itself, each scaled by $1/\varphi^2 \approx 0.382$

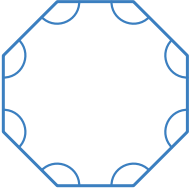
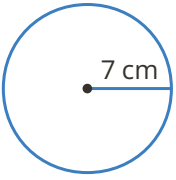
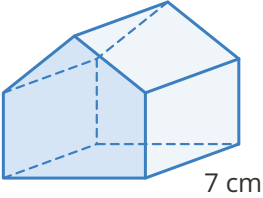
Week 1 · Session 1

TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the sum of the interior angles of this octagon. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 The shaded face has an area of 20 cm^2. Find the volume of the prism. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 Calculate. 35% of \$240	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 Find the value of y when $x = 5$. $y = 3x - 2$	FORMULA _____ SUBSTITUTE _____ ANSWER _____

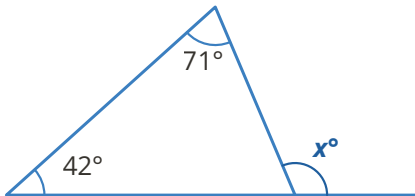
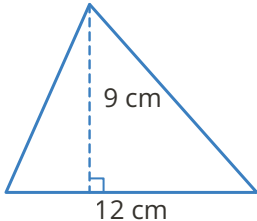
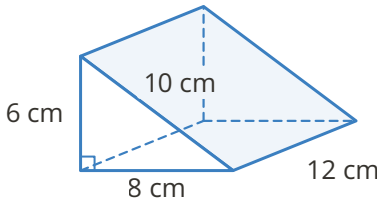
Week 1 · Session 2

TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
2 Find the area of this triangle. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
3 Find the surface area of this triangular prism. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
4 Find the percentage decrease in price. <table border="1" data-bbox="311 1541 601 1666"><tr><td>Original price</td><td>\$80</td></tr><tr><td>New price</td><td>\$68</td></tr></table>	Original price	\$80	New price	\$68	FORMULA _____ SUBSTITUTE _____ ANSWER _____		
Original price	\$80						
New price	\$68						
5 Find the range of these race times. <table border="1" data-bbox="146 1921 783 1986"><tr><td>Seconds</td><td>12.4</td><td>13.1</td><td>11.8</td><td>12.9</td><td>13.6</td></tr></table>	Seconds	12.4	13.1	11.8	12.9	13.6	FORMULA _____ SUBSTITUTE _____ ANSWER _____
Seconds	12.4	13.1	11.8	12.9	13.6		

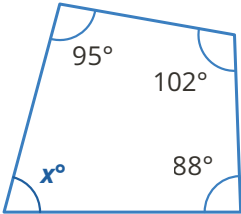
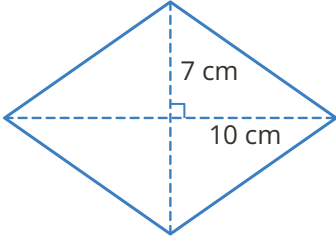
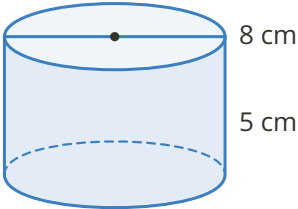
Week 1 · Session 3

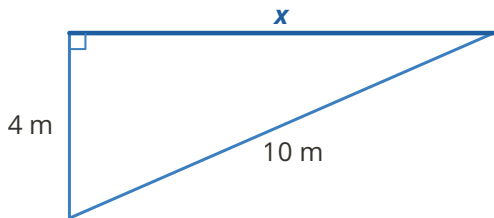
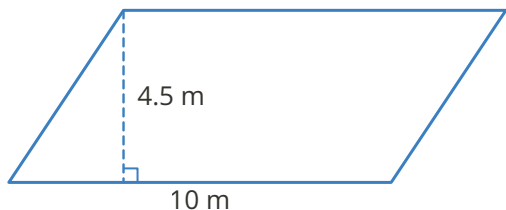
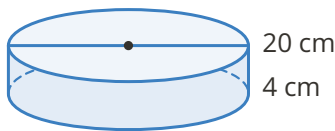
TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING				
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				
2 Find the area of this rhombus. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				
3 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				
4 Find the loss made on this sale. <table border="1" data-bbox="319 1543 596 1668"> <tr> <td>Cost price</td><td>\$85</td></tr> <tr> <td>Selling price</td><td>\$60</td></tr> </table>	Cost price	\$85	Selling price	\$60	FORMULA _____ SUBSTITUTE _____ ANSWER _____
Cost price	\$85				
Selling price	\$60				
5 Find the probability of not winning. $P(\text{win}) = \frac{3}{8}$	FORMULA _____ SUBSTITUTE _____ ANSWER _____				

QUESTION	WORKING						
1 Find x, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
2 Find the area of this parallelogram. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
4 Find the number. 12% of a number is 9.6	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
5 Find the mean of these lengths. <table border="1" data-bbox="146 1919 764 1986"><tr><td>Metres</td><td>2.4</td><td>3.1</td><td>2.8</td><td>3.5</td><td>3.2</td></tr></table>	Metres	2.4	3.1	2.8	3.5	3.2	FORMULA _____ SUBSTITUTE _____ ANSWER _____
Metres	2.4	3.1	2.8	3.5	3.2		

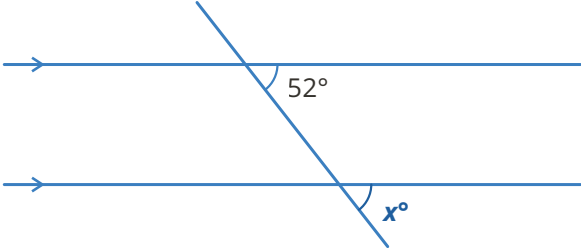
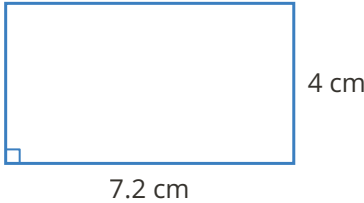
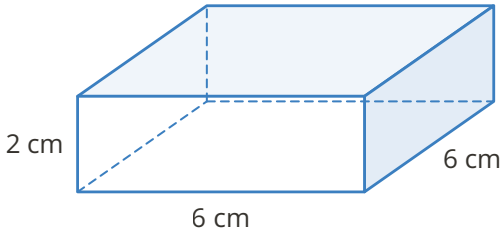
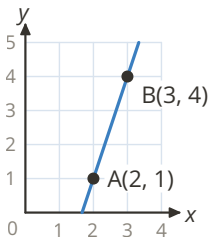
Week 2 · Session 2

TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this rectangle. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 Find the volume of this rectangular prism. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 Expand. $-2(x + 6)$	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 Find the gradient of line AB. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____

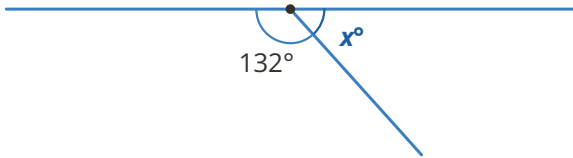
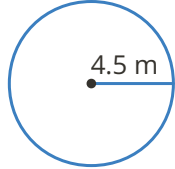
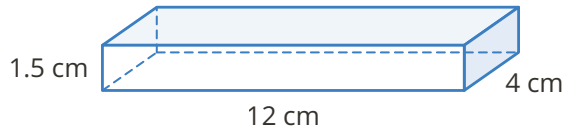
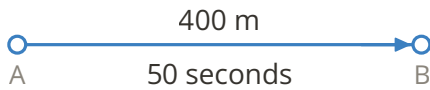
Week 2 · Session 3

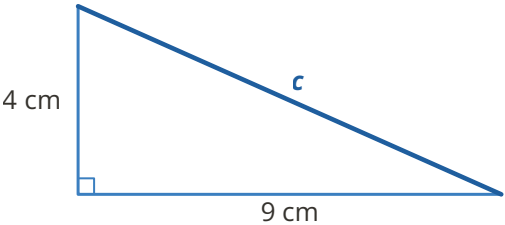
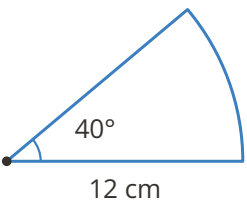
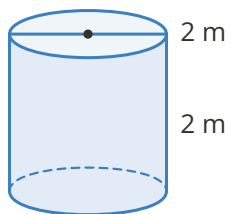
TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
3 Find the surface area of this prism. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
4 A runner travels from A to B. Find the average speed. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
5 A letter is chosen at random from this word. Find $P(A)$. B A N A N A	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____

QUESTION	WORKING				
1 Find c, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				
2 Find the area of this sector, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				
3 Find the capacity of this tank in litres, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				
4 Find the new population of the town. <table border="1" data-bbox="327 1545 590 1668"> <tr> <td>Population</td><td>3,840</td></tr> <tr> <td>Increase</td><td>2.5%</td></tr> </table>	Population	3,840	Increase	2.5%	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
Population	3,840				
Increase	2.5%				
5 Find the value of y when $x = 6$. $y = 0.5x + 4$	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				

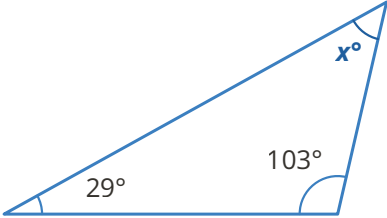
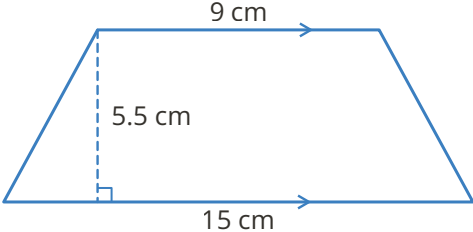
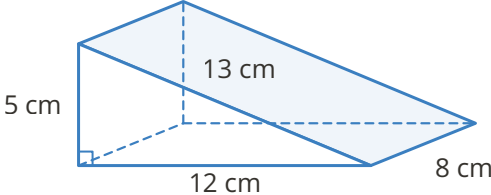
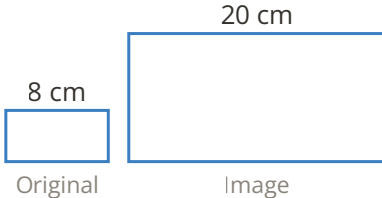
Week 3 · Session 2

TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this trapezium. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this triangular prism. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 The small rectangle is enlarged to make the large one. Find the scale factor. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 Find the probability that the counter is not red. $P(\text{red}) = \frac{5}{12}$	FORMULA _____ SUBSTITUTE _____ ANSWER _____

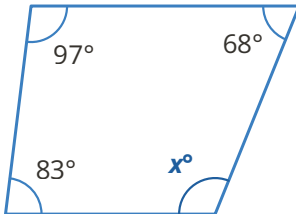
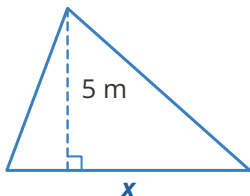
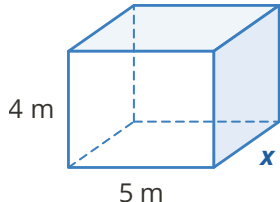
Week 3 · Session 3

TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING								
1 Find the value of x. 	FORMULA SUBSTITUTE ANSWER								
2 The area of this triangle is 18.75 m^2. Find x. 	FORMULA SUBSTITUTE ANSWER								
3 The volume of this rectangular prism is 100 m^3. Find x. 	FORMULA SUBSTITUTE ANSWER								
4 Write as a single power. $10^4 \times 10^3$	FORMULA SUBSTITUTE ANSWER								
5 One marble is chosen at random from this bag. Find the probability that it is green. <table border="1" data-bbox="229 1912 681 2036"><tr><td>Colour</td><td>Red</td><td>Blue</td><td>Green</td></tr><tr><td>Marbles</td><td>4</td><td>2</td><td>6</td></tr></table>	Colour	Red	Blue	Green	Marbles	4	2	6	FORMULA SUBSTITUTE ANSWER
Colour	Red	Blue	Green						
Marbles	4	2	6						

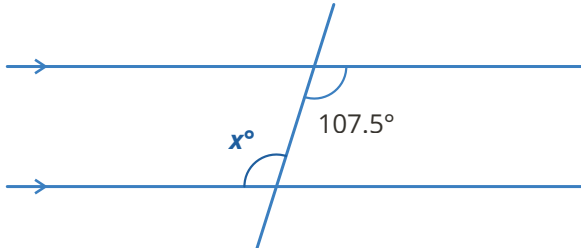
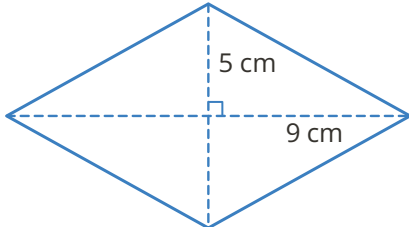
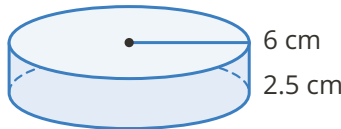
Week 4 · Session 1

TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA SUBSTITUTE ANSWER						
2 Find the area of this rhombus. 	FORMULA SUBSTITUTE ANSWER						
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER						
4 Find the profit made on this sale. <table border="1" data-bbox="301 1541 604 1664"><tr><td>Cost price</td><td>\$12.40</td></tr><tr><td>Selling price</td><td>\$20</td></tr></table>	Cost price	\$12.40	Selling price	\$20	FORMULA SUBSTITUTE ANSWER		
Cost price	\$12.40						
Selling price	\$20						
5 Find the mean of these marks. <table border="1" data-bbox="150 1921 761 1986"><tr><td>Marks</td><td>65</td><td>72</td><td>58</td><td>80</td><td>70</td></tr></table>	Marks	65	72	58	80	70	FORMULA SUBSTITUTE ANSWER
Marks	65	72	58	80	70		

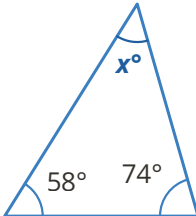
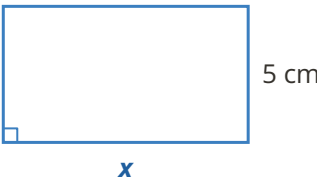
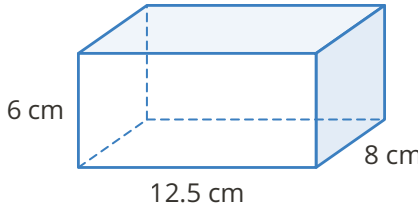
Week 4 · Session 2

TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA SUBSTITUTE ANSWER						
2 The area of this rectangle is 45 cm^2. Find x. 	FORMULA SUBSTITUTE ANSWER						
3 Find the capacity of this container in millilitres. 	FORMULA SUBSTITUTE ANSWER						
4 Find the sale price, to the nearest cent. <table border="1" data-bbox="298 1538 616 1666"><tr><td>Original price</td><td>\$59.99</td></tr><tr><td>Discount</td><td>35%</td></tr></table>	Original price	\$59.99	Discount	35%	FORMULA SUBSTITUTE ANSWER		
Original price	\$59.99						
Discount	35%						
5 Find the range of these scores. <table border="1" data-bbox="150 1919 766 1986"><tr><td>Scores</td><td>35</td><td>48</td><td>29</td><td>52</td><td>41</td></tr></table>	Scores	35	48	29	52	41	FORMULA SUBSTITUTE ANSWER
Scores	35	48	29	52	41		

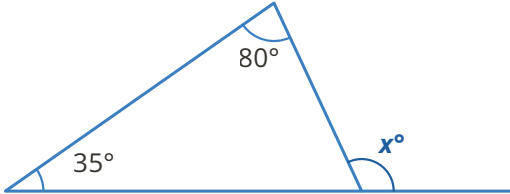
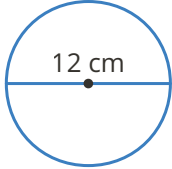
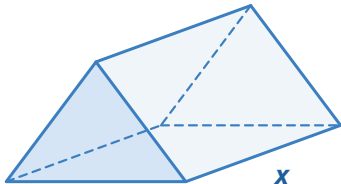
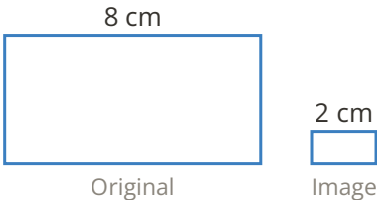
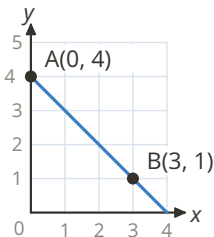
Week 4 · Session 3

TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 The shaded face has an area of 12 cm^2 and the volume is 96 cm^3. Find x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 The large rectangle is reduced to make the small one. Find the scale factor. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 Find the gradient of line AB. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____

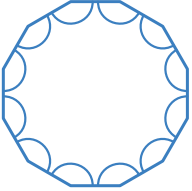
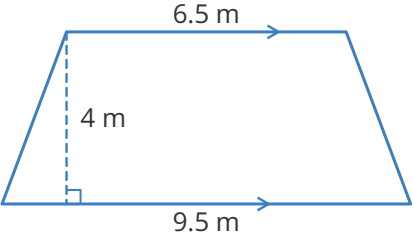
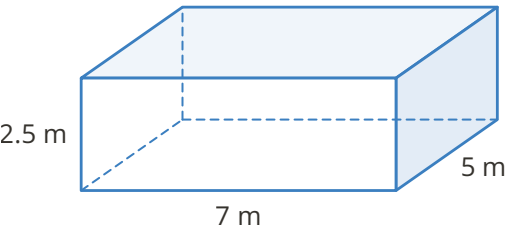
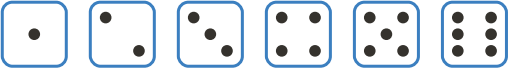
Week 5 · Session 1

TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING				
1 Find the sum of the interior angles of this dodecagon. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				
2 Find the area of this trapezium. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				
3 Find the surface area of this prism. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				
4 Find the percentage increase in visitors. <table border="1" data-bbox="331 1541 582 1668"> <tr> <td>Last year</td><td>1,200</td></tr> <tr> <td>This year</td><td>1,500</td></tr> </table>	Last year	1,200	This year	1,500	FORMULA _____ SUBSTITUTE _____ ANSWER _____
Last year	1,200				
This year	1,500				
5 A fair die is rolled. Find the probability of rolling a number less than 3. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				

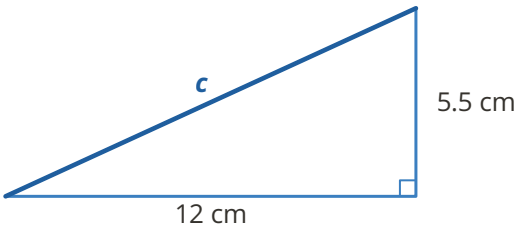
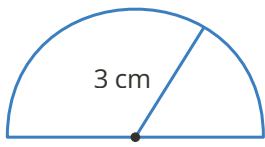
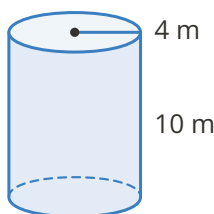
Week 5 · Session 2

TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find c, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER						
2 Find the area of this semicircle, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER						
3 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER						
4 Expand. $2(3x + 5)$	FORMULA SUBSTITUTE ANSWER						
5 Find the mean of these scores. <table border="1" data-bbox="150 1924 764 1986"><tr><td>Scores</td><td>8</td><td>3</td><td>5</td><td>6</td><td>8</td></tr></table>	Scores	8	3	5	6	8	FORMULA SUBSTITUTE ANSWER
Scores	8	3	5	6	8		

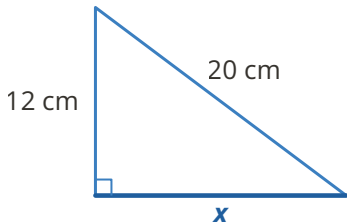
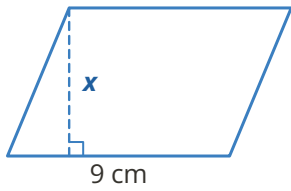
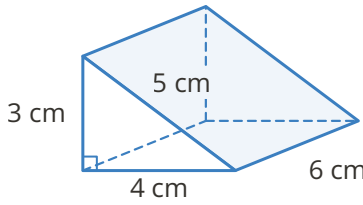
Week 5 · Session 3

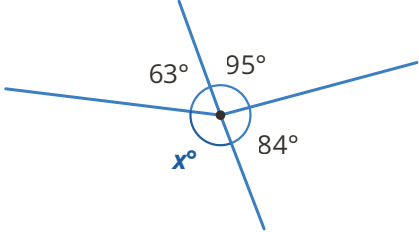
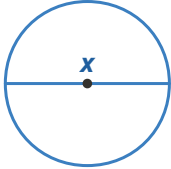
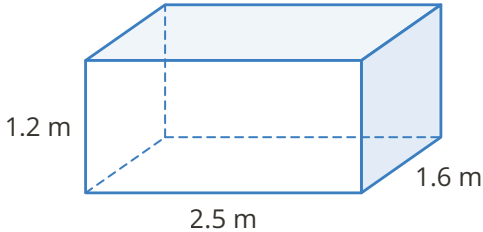
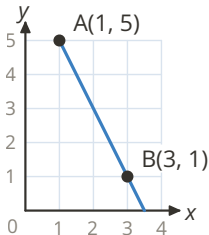
TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
2 The area of this parallelogram is 54 cm^2. Find x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
3 Find the surface area of this triangular prism. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
4 Calculate. 8% of \$65	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
5 Find the range of these heights. <table border="1" data-bbox="150 1919 761 1986"><tr><td>cm</td><td>104</td><td>96</td><td>118</td><td>101</td><td>99</td></tr></table>	cm	104	96	118	101	99	FORMULA _____ SUBSTITUTE _____ ANSWER _____
cm	104	96	118	101	99		

QUESTION	WORKING
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 The circumference of this circle is 100 m. Find x, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 Find the capacity of this tank in litres. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 Simplify. $(3m)^0$	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 Find the gradient of line AB. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____

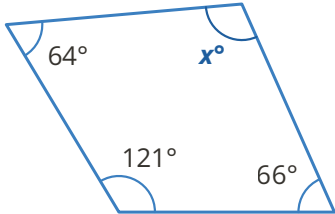
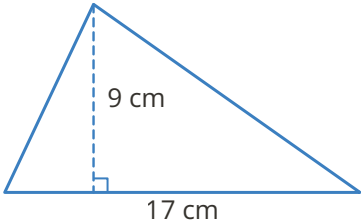
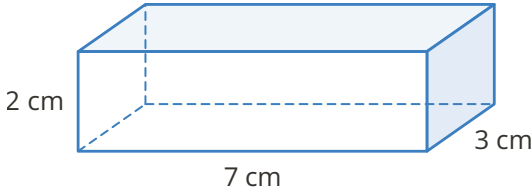
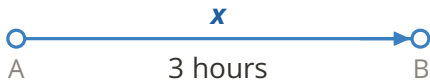
Week 6 · Session 2

TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this triangle. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this prism. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 A car travels from A to B at an average speed of 80 km/h. Find x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 Find the value of y when $x = -1$. $y = -3x + 2$	FORMULA _____ SUBSTITUTE _____ ANSWER _____

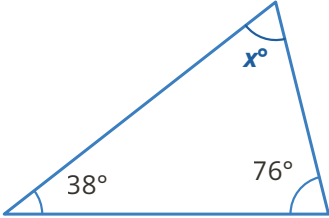
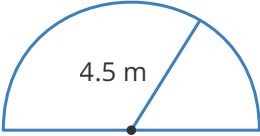
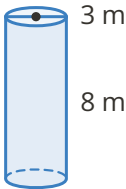
Week 6 · Session 3

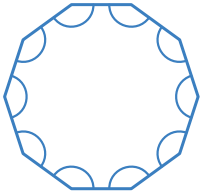
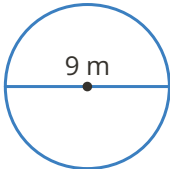
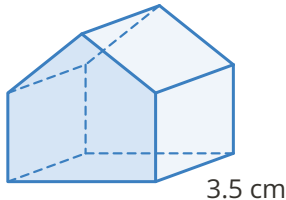
TERM 4

Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING				
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				
2 Find the area of this semicircle, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				
3 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				
4 Find the original price of the jacket. <table border="1" data-bbox="331 1541 582 1668"> <tr> <td>Discount</td><td>20%</td></tr> <tr> <td>Sale price</td><td>\$64</td></tr> </table>	Discount	20%	Sale price	\$64	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
Discount	20%				
Sale price	\$64				
5 Find the probability that the marble is not blue. $P(\text{blue}) = 0.36$	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				

QUESTION	WORKING						
1 Find the size of each interior angle of this regular decagon. 	FORMULA SUBSTITUTE ANSWER						
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER						
3 The shaded face has an area of 20 cm^2. Find the volume of the prism. 	FORMULA SUBSTITUTE ANSWER						
4 Calculate. 2.5% of 480 m	FORMULA SUBSTITUTE ANSWER						
5 Find the mean of these times. <table border="1" data-bbox="150 1921 759 1986"><tr><td>Hours</td><td>1.5</td><td>2.5</td><td>3</td><td>2</td><td>1</td></tr></table>	Hours	1.5	2.5	3	2	1	FORMULA SUBSTITUTE ANSWER
Hours	1.5	2.5	3	2	1		

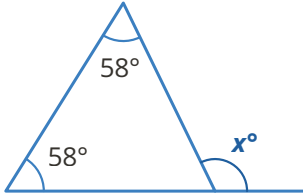
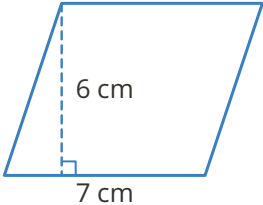
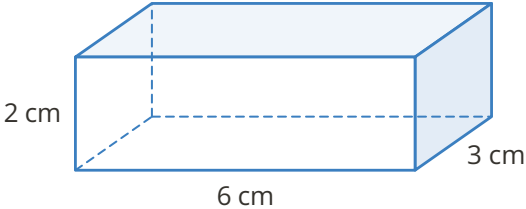
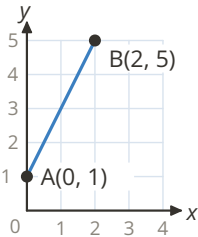
Week 7 · Session 2

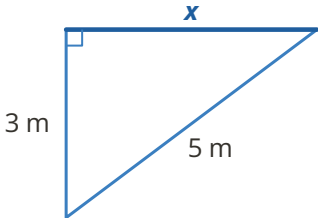
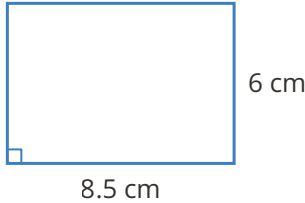
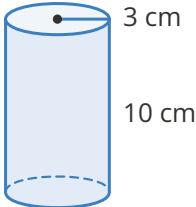
TERM 4

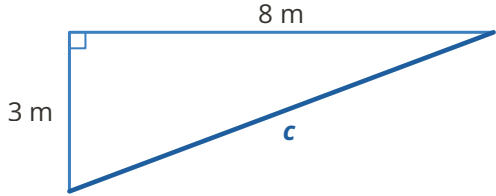
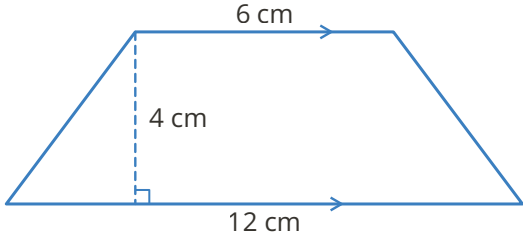
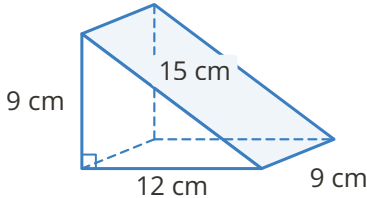
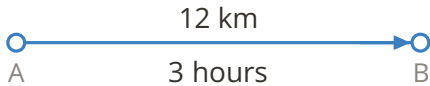
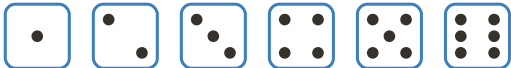
Write the formula, substitute, then solve.

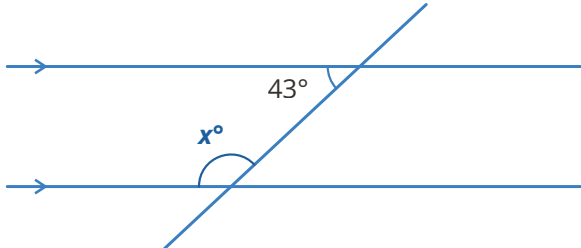
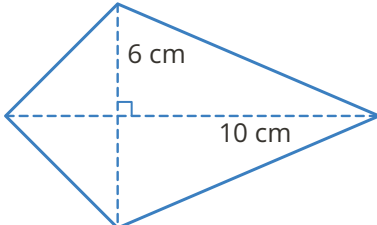
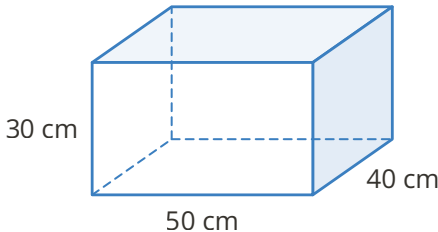
NAME _____

DATE _____

QUESTION	WORKING				
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				
2 Find the area of this parallelogram. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				
3 Find the volume of this rectangular prism. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				
4 Find the percentage decrease in price. <table border="1" data-bbox="311 1543 603 1666"> <tr> <td>Original price</td><td>\$160</td></tr> <tr> <td>New price</td><td>\$136</td></tr> </table>	Original price	\$160	New price	\$136	FORMULA _____ SUBSTITUTE _____ ANSWER _____
Original price	\$160				
New price	\$136				
5 Find the gradient of line AB. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				

QUESTION	WORKING				
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				
2 Find the area of this rectangle. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				
4 Find the original price of the jacket. <table border="1" data-bbox="323 1541 590 1668"> <tr> <td>Discount</td><td>12.5%</td></tr> <tr> <td>Sale price</td><td>\$70</td></tr> </table>	Discount	12.5%	Sale price	\$70	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
Discount	12.5%				
Sale price	\$70				
5 Find the probability that the team does not win. $P(\text{win}) = 0.35$	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				

QUESTION	WORKING
1 Find c, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this trapezium. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this triangular prism. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 A walker travels from A to B. Find the average speed. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 A fair die is rolled. Find the probability of rolling a number greater than 1. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA SUBSTITUTE ANSWER						
2 Find the area of this kite. 	FORMULA SUBSTITUTE ANSWER						
3 Find the capacity of this tank in litres. 	FORMULA SUBSTITUTE ANSWER						
4 Simplify. $k^{10} \div k^4$	FORMULA SUBSTITUTE ANSWER						
5 Find the range of these temperatures. <table border="1" data-bbox="150 1921 759 1984"><tr><td>°C</td><td>-8</td><td>-2</td><td>-5</td><td>3</td><td>1</td></tr></table>	°C	-8	-2	-5	3	1	FORMULA SUBSTITUTE ANSWER
°C	-8	-2	-5	3	1		

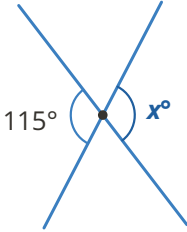
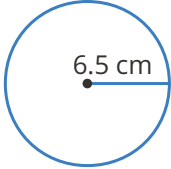
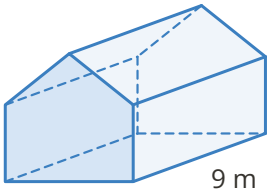
Week 8 · Session 3

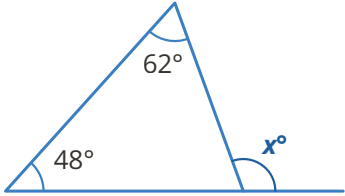
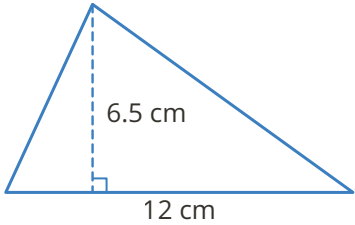
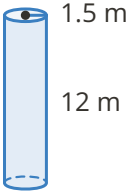
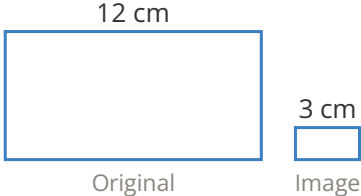
TERM 4

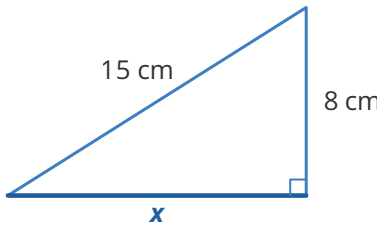
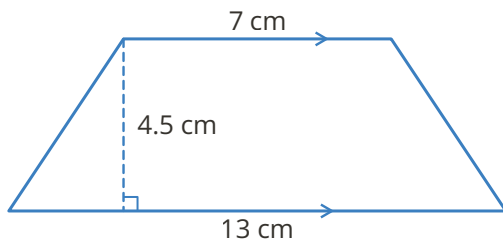
Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING				
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				
3 The shaded face has an area of 20 m^2. Find the volume of the prism. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				
4 Find the new price. <table border="1" data-bbox="300 1541 616 1668"> <tr> <td>Original price</td><td>\$64.50</td></tr> <tr> <td>Increase</td><td>12%</td></tr> </table>	Original price	\$64.50	Increase	12%	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
Original price	\$64.50				
Increase	12%				
5 Find the value of y when $x = 4$. $y = 3x + 1$	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____				

QUESTION	WORKING
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this triangle. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 The large rectangle is reduced to make the small one. Find the scale factor. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 Find the probability that the target is not hit. $P(\text{hit}) = \frac{7}{9}$	FORMULA _____ SUBSTITUTE _____ ANSWER _____

QUESTION	WORKING						
1 Find x, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
2 Find the area of this trapezium. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
3 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
4 Expand. $6(2a - 1)$	FORMULA _____ SUBSTITUTE _____ ANSWER _____						
5 Find the mean of these masses. <table border="1" data-bbox="150 1921 761 1986"><tr><td>kg</td><td>9.5</td><td>8</td><td>10.5</td><td>12</td><td>7</td></tr></table>	kg	9.5	8	10.5	12	7	FORMULA _____ SUBSTITUTE _____ ANSWER _____
kg	9.5	8	10.5	12	7		

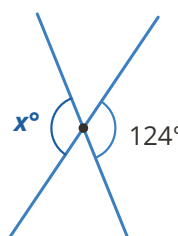
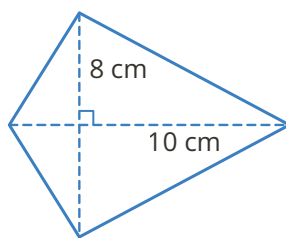
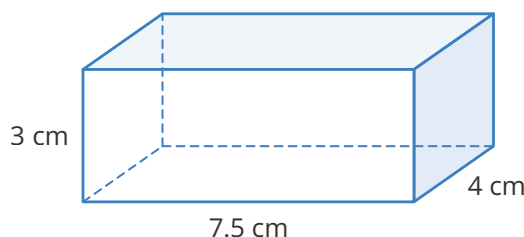
Week 9 · Session 3

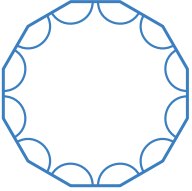
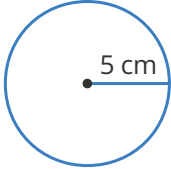
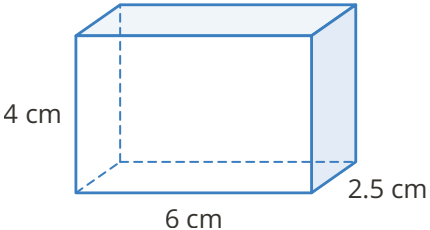
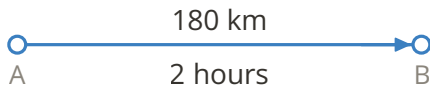
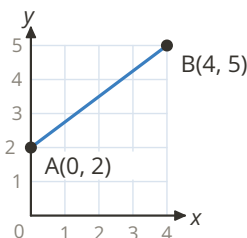
TERM 4

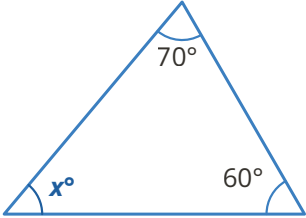
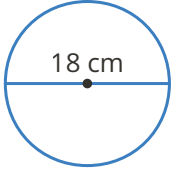
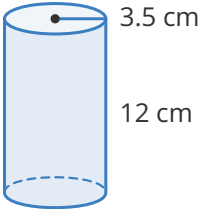
Write the formula, substitute, then solve.

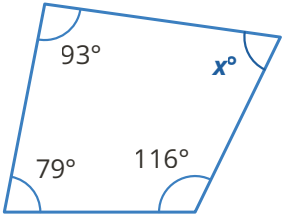
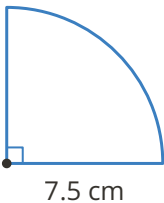
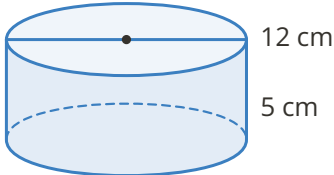
NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA SUBSTITUTE ANSWER						
2 Find the area of this kite. 	FORMULA SUBSTITUTE ANSWER						
3 Find the volume of this rectangular prism. 	FORMULA SUBSTITUTE ANSWER						
4 Find the loss made on this sale. <table border="1" data-bbox="301 1532 603 1666"><tr><td>Cost price</td><td>\$65</td></tr><tr><td>Selling price</td><td>\$52.75</td></tr></table>	Cost price	\$65	Selling price	\$52.75	FORMULA SUBSTITUTE ANSWER		
Cost price	\$65						
Selling price	\$52.75						
5 Find the range of these heights. <table border="1" data-bbox="140 1912 764 1980"><tr><td>Metres</td><td>1.62</td><td>1.48</td><td>1.75</td><td>1.53</td><td>1.69</td></tr></table>	Metres	1.62	1.48	1.75	1.53	1.69	FORMULA SUBSTITUTE ANSWER
Metres	1.62	1.48	1.75	1.53	1.69		

QUESTION	WORKING
1 Find the size of each interior angle of this regular dodecagon. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
3 Find the surface area of this prism. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
4 A car travels from A to B. Find the average speed. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
5 Find the gradient of line AB. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 Find the capacity of this container in millilitres, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 Simplify. $(n^3)^4$	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 A fair die is rolled. Find the probability of rolling an even number. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____

QUESTION	WORKING				
1 Find the value of x. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				
2 Find the area of this quadrant, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____				
4 Find the loss made on this sale. <table border="1" data-bbox="306 1541 609 1668"> <tr> <td>Cost price</td><td>\$84.50</td></tr> <tr> <td>Selling price</td><td>\$70</td></tr> </table>	Cost price	\$84.50	Selling price	\$70	FORMULA _____ SUBSTITUTE _____ ANSWER _____
Cost price	\$84.50				
Selling price	\$70				
5 Find the value of y when $x = 3$. $y = -2x + 10$	FORMULA _____ SUBSTITUTE _____ ANSWER _____				